Social and technological drivers of sustainable manufacturing performance: A cognitive-organizational perspective

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Abstract: Sustainable manufacturing has become a strategic priority for organizations seeking to balance competitiveness with environmental and social responsibility. However, the successful implementation of sustainability initiatives depends not only on technological and economic factors but also on workforce competencies, organizational learning, and human-centered drivers. This study examines the cognitive and organizational determinants of sustainable manufacturing performance through the analysis of social, economic, environmental, and quality drivers in manufacturing organizations in the Republic of Serbia. Data were collected using a structured questionnaire administered to 153 employees in production-related positions. The research model integrates human, technological, and managerial dimensions of sustainability and evaluates their influence on six performance outcomes: environmental, social, economic, process, innovation, and stakeholder performance. Multiple regression analyses were applied to test the proposed relationships. The findings indicate that environmental and economic drivers exert the strongest positive effects on performance outcomes, while social drivers - reflecting employee engagement, training, and organizational culture - demonstrate moderate but significant influence. Quality drivers show mixed effects across performance domains. The results highlight the critical role of workforce involvement, knowledge development, and sustainability-oriented competencies in supporting organizational transformation toward sustainable manufacturing. This study contributes to the emerging cognitive-organizational perspective on sustainability by integrating human and technological drivers within a single empirical framework and providing evidence from a transition economy context.

Keywords: sustainable manufacturing; cognitive-organizational drivers; organizational performance; workforce competencies; technological drivers; human factors

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Introduction

Sustainable manufacturing is evolving as organizations adopt strategies to reduce waste, enhance efficiency, and align competitiveness with environmental and social responsibility (Karuppiah et al., 2024; Scharmer et al., 2024). The concept of sustainable manufacturing encompasses the integration of environmental management, social responsibility, and operational excellence into production processes to achieve long-term viability (Scharmer et al., 2024). However, despite theoretical advances, the empirical

evidence on which drivers are crucial in the process of implementing sustainable manufacturing, especially in transitional economies, remains limited. Companies are increasingly under pressure to align competitiveness with responsibility toward the environment, employees, and society. Beyond technological investments and environmental initiatives, sustainable manufacturing increasingly depends on human and cognitive factors embedded within organizational systems. Employee competencies, sustainability awareness, continuous learning, and knowledge sharing practices shape the extent to which sustainability principles are adopted and operationalized in production environments. From this perspective, manufacturing transformation is not solely a technical process but also a cognitive and educational one, requiring the development of workforce capabilities aligned with sustainable and digital industrial paradigms. In transition economies, the pressure rises, as traditional industrial models must adapt to the requirements of sustainability and digital transformation (Karuppiah et al., 2024

1.1 Literature review

Achieving sustainability-oriented performance improvement requires a multidimensional approach that integrates organizational, technological, and human factors (Rosen & Kishawy, 2012). The idea has evolved in recent years from being solely ecological to encompassing environmental management in relation to innovation and operational excellence (Andronie et al., 2021). Manufacturing businesses are increasingly recognizing that sustainability is not just a moral or environmental requirement; it is also a business imperative. Resilience and long-term competitiveness are fueled by sustainability (Andronie et al., 2021). The internal and external factors that are essential for organizations to implement socially and environmentally responsible practices are represented by drivers in this framework. These drivers are typically divided into four groups based on a summary of earlier research: social, economic, environmental, and quality drivers. Each of these groups influences sustainability outcomes through specific mechanisms. These factors frequently work together to produce synergies that increase the overall impact on the organization. Social drivers underscore the importance of organizational culture, employee well-being, and human capital as the cornerstones of long-term success. Strong ergonomic work practices and a safety culture increase dependability, lower accident rates, and promote long-term results (Małysa & Gajdzik, 2020).

Materials and methods

To investigate the impact of various factors on sustainable manufacturing performance in production-oriented organizations in Serbia, this study employed a quantitative, cross-sectional research design. This method was considered suitable for evaluating multiple relationships simultaneously and identifying trends across a wide range of industrial settings.

2.1 Data collection

To ensure widespread participation, a structured questionnaire was disseminated both electronically and in print to gather data. Employees from manufacturing companies across various industrial sectors, including food production, machinery, metal processing, and automotive components, provided a total of 153 valid responses. Because the respondents held a variety of professional roles, from mana-

gerial and administrative to technical and operational, the sample was representative of those directly involved in or supporting production processes

2.2 Sample characteristics

The final sample consisted of 153 respondents employed in manufacturing and production-related organizations in the Republic of Serbia. The respondents' age ranged from 20 to 66 years (M = 41.19, SD = 8.74), indicating a mature and professionally experienced sample. With respect to gender, 140 respondents were male (91.5%) and 13 were female (8.5%), reflecting the gender structure commonly observed in manufacturing and industrial sectors. The demographic characteristics of the respondents are summarised in Table 1. The sample is predominantly male (90.8%), with most respondents holding a university degree (75.8%).

Table 1 *Demographic characteristics of respondents (N=153).*

Variable	Category	n	%
Gender	Male	139	90.8
	Female	14	9.2
Education level	Secondary school	10	6.5
	College	19	12.4
	University degree	116	75.8
	Doctoral degree	10	6.5
Age (years)	Mean (SD)	41.2 (9.8)	
	Range	20–66	

2.3 Instrument development and data analysis

Prior empirical studies on organizational performance and sustainable manufacturing provided the foundation for developing the questionnaire. It contained measurement items intended to evaluate four types of drivers: quality drivers (QM), environmental protection drivers (EPM), economic drivers (EM), and social drivers (SM). Environmental performance (ENP), social performance (SP), economic performance (EP), process performance (PP), innovation and development performance (IDP), and stakeholder performance (STP) were the six performance outcomes that were measured accordingly.

Results

3.1 Descriptive statistics and correlations between study variables

Descriptive statistics, reliability coefficients, and correlations between study variables are presented in Table 2. All scales demonstrated acceptable internal consistency, with Cronbach's alpha values ranging from .69 to .87. The distributions of the variables were approximately normal, with skewness and kurtosis values within recommended thresholds (±2). The four types of drivers were positively and

strongly correlated with each other (rs = .57–.74, ps < .001). They also showed positive associations with all performance outcomes, with correlations ranging from moderate to strong (rs = .41–.75, ps < .001). The pattern of correlations suggests that stronger motivational orientations are consistently associated with higher performance across environmental, social, economic, process, innovation, development, and stakeholder domains. Among the performance indicators, intercorrelations were consistently strong (rs = .51–.76, ps < .001), indicating that higher levels of one type of performance were generally accompanied by higher levels of other performance dimensions.

Table 2 *Descriptive statistics, reliability coefficients, and correlations between study variables.*

	1	2	3	4	5	6	7	8	9	10
1. Social drivers	1									
2. Economic drivers	0.622 **	1								
3. Environmental drivers	0.624 **	0.734 **	1							
4. Quality drivers	0.628 **	0.643 **	0.571 **	1						
5. Environmental performance	0.524 **	0.611 **	0.704 **	0.445 **	1					
6. Social performance	0.559 **	0.645 **	0.604 **	0.579 **	0.619 **	1				
7. Economic performance	0.592 **	0.633 **	0.629 **	0.586 **	0.704 **	0.701 **	1			
8. Process performance	0.557 **	0.624 **	0.660 **	0.558 **	0.655 **	0.668 **	0.746 **	1		
9. Innovation and development performance	0.474 **	0.638 **	0.665 **	0.417 **	0.751 **	0.647 **	0.764 **	0.732 **	1	
10. Stakeholder performance	0.494 **	0.552 **	0.500 **	0.528 **	0.508 **	0.630 **	0.687 **	0.701 **	0.608 **	1
Min–max	6–25	4–19	4–20	4–20	5–24	4–20	4–20	4–20	4–20	4–20
M	18.58	14.77	14.75	15.05	17.98	15.21	14.13	14.95	14.98	14.86
SD	3.36	2.79	3.29	3.01	3.99	2.94	3.19	2.79	3.34	3.08
Skewness	−0.85	−1.34	−1.27	−1.06	−1.10	−1.23	−0.82	−1.39	−1.21	−1.18
Kurtosis	0.86	2.55	1.87	1.12	0.69	1.91	0.88	3.13	1.30	2.06
α	0.69	0.71	0.74	0.75	0.79	0.76	0.76	0.74	0.87	0.81

Note: M = mean; SD = standard deviation. ** p < 0.01.

3.2. Multiple regression analyses of motivational factors predicting performance outcomes

Results of six multiple regression analyses are presented in Table 3. Together, the four types of drivers explained between 36% and 51% of the variance in performance outcomes, with adjusted R^2 values ranging from .36 (stakeholder performance) to .51 (environmental performance). For environmental performance, environmental drivers were the strongest positive predictor (β = .53, p < .01), and economic drivers also showed a significant positive effect (β = .19, p < .05), whereas social and quality drivers were nonsignificant. In the case of social performance, three drivers, economic (β = .30, p < .01), environmental (β = .19, p < .05), and quality (β = .20, p < .05), emerged as significant positive predictors, while social drivers did not contribute significantly. For economic performance, all four drivers showed significant positive associations: social (β = .18, p < .05), economic (β = .21, p < .05), environmental (β = .25, p < .01), and quality (β = .19, p < .05). In contrast, for process performance, only environmental drivers significantly predicted outcomes (β = .36, p < .01), while the remaining predictors were nonsignificant. Regarding innovation and development performance, economic (β = .35, p < .01) and environmental drivers (β = .43,

p < .01) both emerged as strong predictors, whereas social and quality drivers did not reach significance. Finally, for stakeholder performance, economic (β = .25, p < .05) and quality drivers (β = .23, p < .05) were significant positive predictors, while social and environmental drivers were nonsignificant. Taken together, the results indicate that environmental and economic drivers most consistently predicted performance across domains, with environmental drivers in particular showing the strongest effects for environmental and innovation and development outcomes.

Table 3 *Multiple regression analyses predicting performance outcomes from motivational factors.*

	Performance											
	Environmental		Social		Economic		Process		IDP		Stakeholder	
Drivers	**B (SE)**	**β**	**B (SE)**	**β**	**B (SE)**	**B**	**B (SE)**	**β**	**B (SE)**	**β**	**B (SE)**	**β**
Social	0.13 (0.09)	0.11	0.12 (0.07)	0.13	0.18 (0.08)	0.18 *	0.10 (0.07)	0.16	0.04 (0.08)	0.04	0.12 (0.08)	0.13
Economic	0.27 (0.13)	0.19 *	0.31 (0.10)	0.30 **	0.24 (0.11)	0.21 *	0.18 (0.09)	0.18	0.42 (0.11)	0.35 **	0.27 (0.12)	0.25 *
Environmental	0.64 (0.11)	0.53 **	0.17 (0.08)	0.19 *	0.24 (0.09)	0.25 **	0.31 (0.08)	0.36 **	0.44 (0.09)	0.43 **	0.10 (0.09)	0.11
Quality	−0.06 (0.11)	−0.04	0.19 (0.08)	0.20*	0.21 (0.09)	0.19 *	0.15 (0.08)	0.16	0.08 (0.09)	0.08	0.23 (0.09)	0.23 *
R^2	0.521		0.494		0.512		0.507		0.494		0.374	
adj. R^2	0.508		0.480		0.499		0.494		0.480		0.357	

Note. B = unstandardized regression coefficient (standard errors in parentheses); β = standardized regression coefficient; R2 = coefficient of determination; adj. R2 = adjusted R2. * p < 0.05; ** p < 0.01.

Discussion

The study's findings offer new empirical insights into how social, economic, environmental, and quality factors impact long-term organizational performance in Serbia's manufacturing industry. The results indicate that social drivers have a moderate but positive influence, while quality drivers exhibit mixed effects. In contrast, economic and environmental drivers have the most substantial impact on performance outcomes, aligning with global research trends. The significant influence of economic drivers underscores the ongoing importance of cost optimization, efficiency, and innovation in fostering sustainable manufacturing. Technologies such as automation, artificial intelligence, and predictive maintenance can increase productivity while reducing waste production and operating expenses (Polese et al., 2021; Mahfoud et al., 2024; Agrawal et al., 2023).

Conclusions

This research examined how quality, social, economic, and environmental factors influence long-term organizational performance in Serbian manufacturing firms. The findings underscore the importance of creativity, effectiveness, and ecological responsibility in promoting sustainability, demonstrating that both economic and environmental factors have a significant positive impact. By emphasizing

the growing importance of employee engagement, safety, and well-being in creating resilient organizations, social drivers also play a significant role.

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Conflict of interests

The authors declare no conflict of interest.

Data availability statement

The original contributions presented in the study are included in the article. Further inquiries can be directed to the corresponding author.

Institutional Review Board Statement

All ethical considerations were observed, including the confidentiality of personal information, voluntary participation, and data protection in accordance with applicable legal and institutional guidelines.

Author Contributions

Conceptualization, S.Vasić; methodology, S.Vasić; software, S.Vasić. and K.R.; validation, S.Vasić and N.B.; formal analysis, S.Vasić; investigation, S.Vasić and N.B.; resources, S.Vulanović and N.B.; data curation, S.Vasić., K.R., D.V. and N.B.; writing-original draft preparation, S.Vasić; writing-review and editing, S.Vasić.; visualization, S.Vulanović; supervision, N.B.; project administration, S.Vasić and K.R.; funding acquisition, S.Vasić, K.R., N.B., and D.V.;

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